Systems Analysis and Control

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Lecture 16: Lead-Lag Controllers

Overview

In this Lecture, you will learn:

Compensation via Root-Locus

- Introduction
- Pole-Zero Compensation
- Lead-Lag

Can we Add Poles and Zeros to the Root Locus?

Root Locus isn't perfect

- Can only study one parameter at a time.
- What to do if root locus doesn't go where we want?

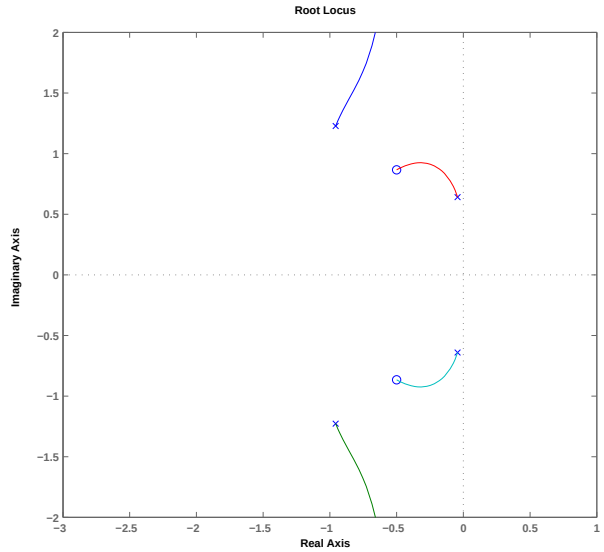
The Suspension Problem:

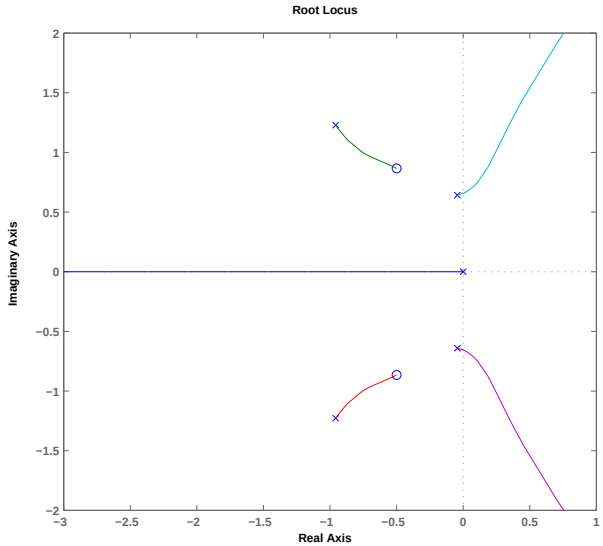
$$G(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + s + 1}$$

Adding a pole at the origin has a negative effect.

Question:

Would adding a zero have a positive effect?





PD Control Adds a Zero to the Root Locus

The Inverted Pendulum:

$$G(s) = \frac{1}{s^2 + .5}$$

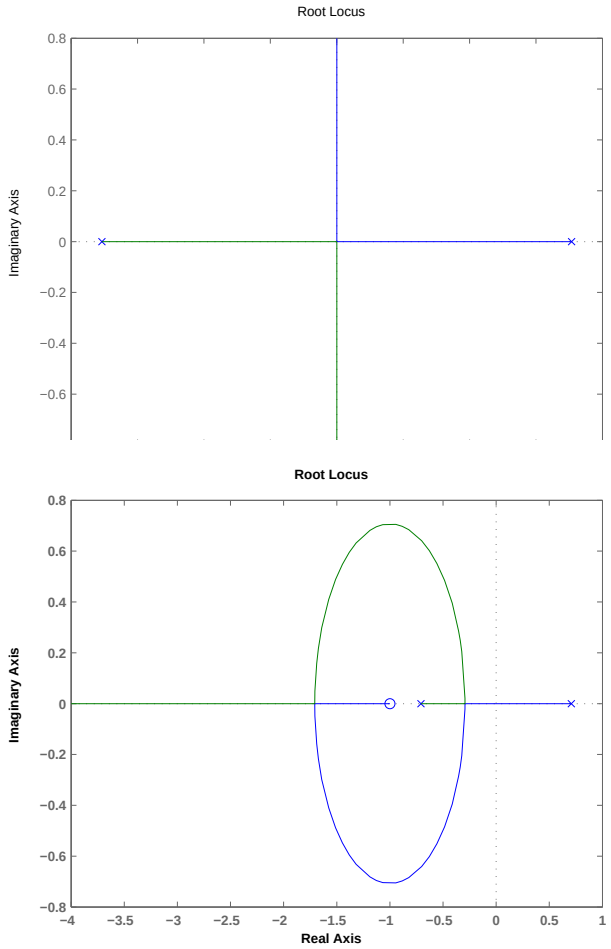
We used PD feedback $K(s) = k(1 + T_D s)$

- Puts a zero at $s = \frac{1}{T_D}$

Conclusion:

- Adding a zero at $z = -1$ improves performance.

Can we generalize this?



PID Control Adds Two Zeros and One Pole (at the origin)

How to choose the locations?

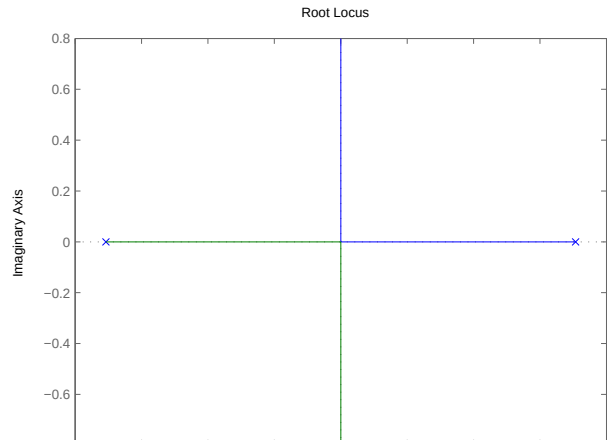
PID feedback:

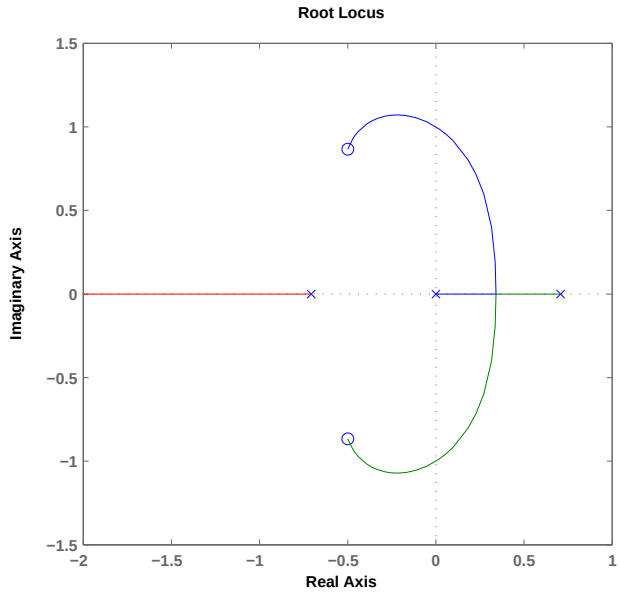
$$K(s) = k\left(1 + T_i\frac{1}{s} + T_D s\right)$$
$$= k\frac{T_D s^2 + s + T_I}{s}$$

Adds poles and zeros:

- Two zeros: $z_{1,2} = -\dfrac{1 \pm \sqrt{1 - 4T_D T_I}}{2T_D}$
- One pole: $p = 0$

Question: Is there a systematic way to add poles and zeros?



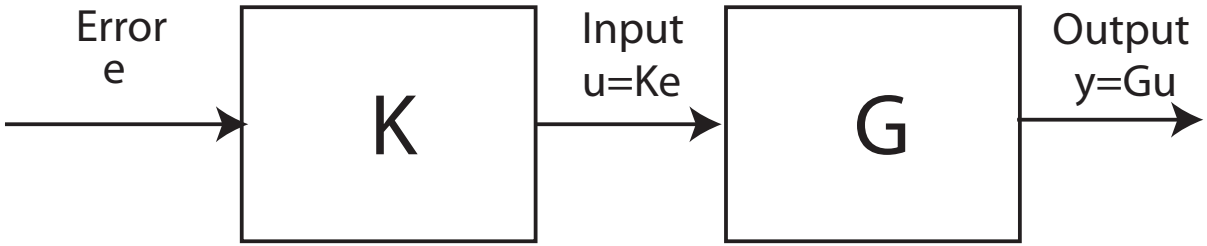


Controllers Can Be Used to Add Poles and Zeros

How?

How To Add a Pole/Zero?

- What does it mean?



Add a pole and Zero? The obvious choice is

$$K(s) = k\frac{s+z}{s+p}$$

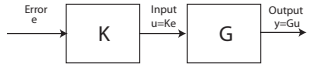
Note: This form isn't normalized!! To add a pole-zero pair without changing the gain, use the equivalent form

$$K(s) = k\frac{s/z+1}{s/p+1}$$

Question: How to implement the controller.

How To Add a Pole/Zero?

- What does it mean?



Add a pole and Zero? The obvious choice is

$$K(s) = k\frac{s+z}{s+p}$$

Note: This form isn't normalized!! To add a pole-zero pair without changing the gain, use the equivalent form

$$K(s) = k\frac{s/z+1}{s/p+1}$$

Question: How to implement the controller.

The importance of not changing the gain is that when we calculate the gain for a point on the modified root locus, it will be for a controller of form

$$K(s) = k\frac{s/z+1}{s/p+1}$$

We can convert back to traditional form using

$$K(s) = k\frac{s/z+1}{s/p+1} = k\frac{p}{z}\frac{s+z}{s+p}$$

How Do Controllers Add Zeros?

Physical Interpretation and Implementation

Goal: Add a Zero

- Just like PD control.

Controller:

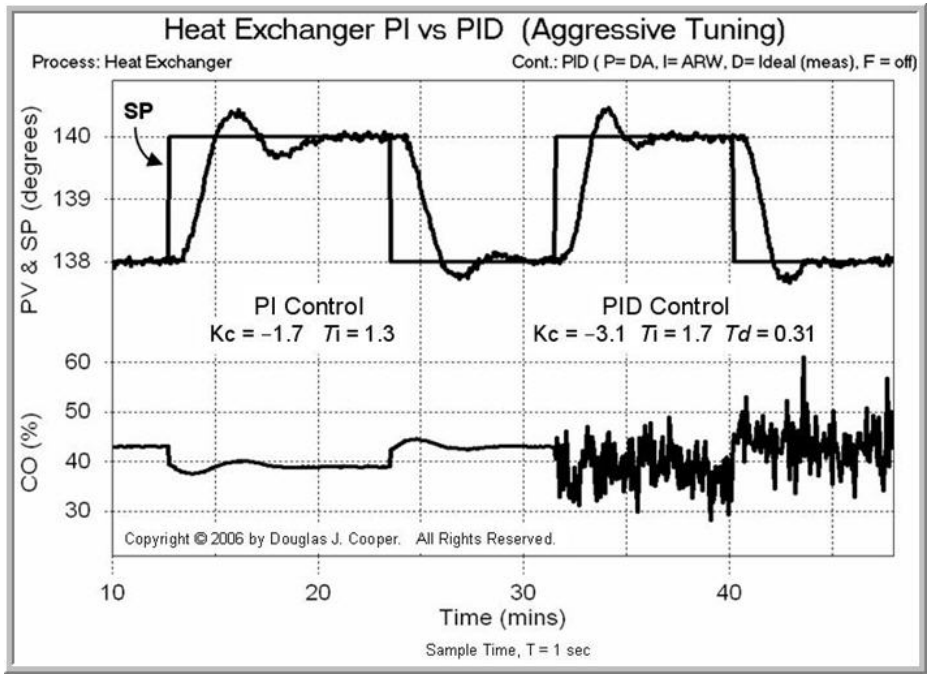
$$K(s) = k(s + z)$$

- **Input to Controller:** $\hat{e}(s)$
- **Output from Controller:**

$$\begin{aligned}\hat{u}(s) &= k(s + z) \\ &= ks\hat{e}(s) + kz\hat{e}(s)\end{aligned}$$

Time-Domain:

$$u(t) = k\, e'(t) + kz\, e(t)$$



Problem: Requires differentiation.

$$e'(t) \cong \frac{e(t) - e(t - \tau)}{\tau}$$

How Do Controllers Add Poles?

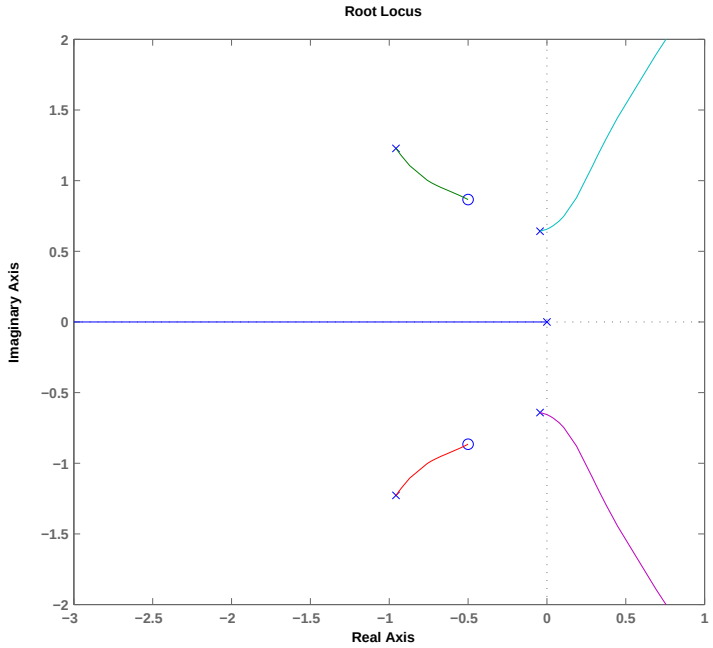
Physical Interpretation and Implementation

Goal: Add a pole

Controller:

$$K(s) = k\frac{1}{s+p}$$

Input to Controller: $\hat{e}(s)$
Output from Controller: $\hat{u}(s) = \frac{k}{s+p}\hat{e}(s)$
Internal Variable: x.



- Frequency Domain

$$(s+p)x(s) = ke(s)$$
$$u(s) = x(s)$$

- Time-Domain

$$\dot{x}(t) = -px(t) + ke(t)$$
$$u(t) = x(t)$$

Adding a Pole:

- Requires us to construct a dynamical system whose output is the control.
- Easier than adding a zero, but less useful

How Do Controllers Add Pole-Zero Pairs?

Use A Simple Gain in Parallel with a Pole

The best way to modify the root locus is a using a pole-zero compensator.

- Adds a zero without differentiation

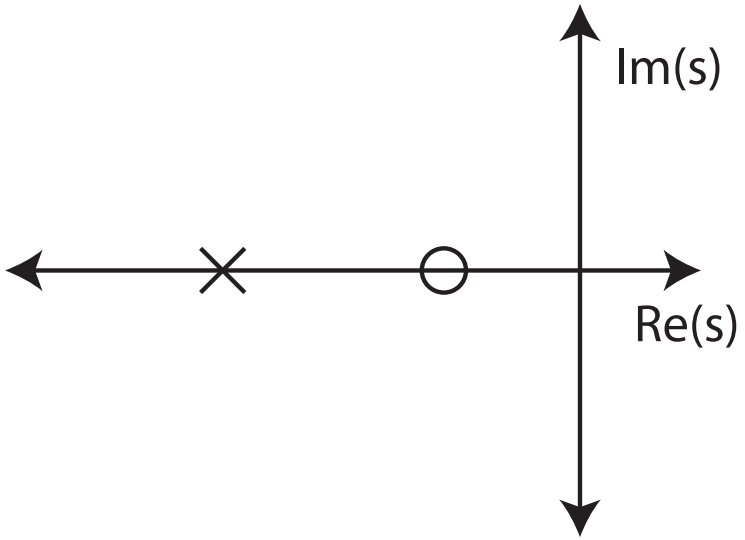
$$K(s) = k\frac{s-z}{s-p}$$

Input to Controller: $\hat{e}(s)$

Output from Controller: $\hat{u}(s) = k\frac{s-z}{s-p}\hat{e}(s)$

Doing long division:

$$\frac{s-z}{s-p} = 1 + \frac{z-p}{s-p}$$



Hence

$$\hat{u}(s) = k\hat{e}(s) + k\frac{z-p}{s-p}\hat{e}(s)$$

Internal Variable:

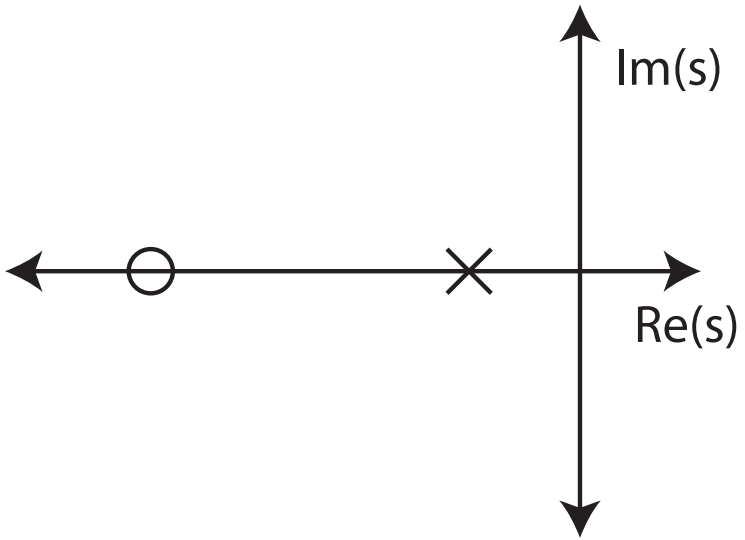
$$\hat{x}(s) = \frac{k(z-p)}{s-p}\hat{e}(s)$$

How Do Controllers Add Pole-Zero Pairs?

Internal Controller Dynamics

Internal Variable:

$$\hat{x}(s) = \frac{k(z-p)}{s-p}\hat{e}(s)$$



- Frequency Domain:

$$(s+p)x(s) = k(z-p)e(s)$$
$$u(s) = x(s) + k\hat{e}(s)$$

- Time-Domain:

$$\dot{x}(t) = -px(t) + k(z-p)e(t)$$
$$u(t) = x(t) + ke(t)$$

Artificial Dynamics:

- Controller State: $x(t)$
- No differentiation of $e(t)$!
- A zero should always be combined with a pole.

Lead Compensation

Definition 1.

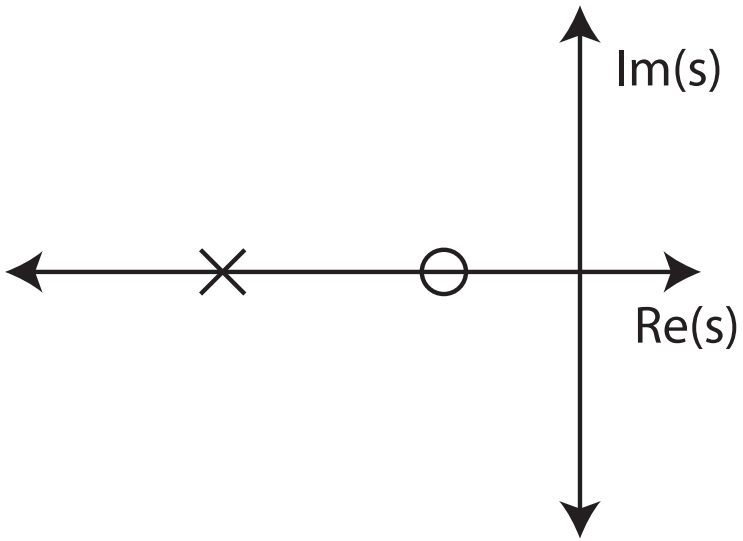
A **Lead Compensator** is a pole-zero compensator

$$K(s) = \frac{s+z}{s+p}$$

where $p < z$.

Used when we really want a zero

- The pole has less effect than the zero.



Lead Compensation

Inverted Pendulum

$$G(s) = \frac{1}{s^2 - .5}$$

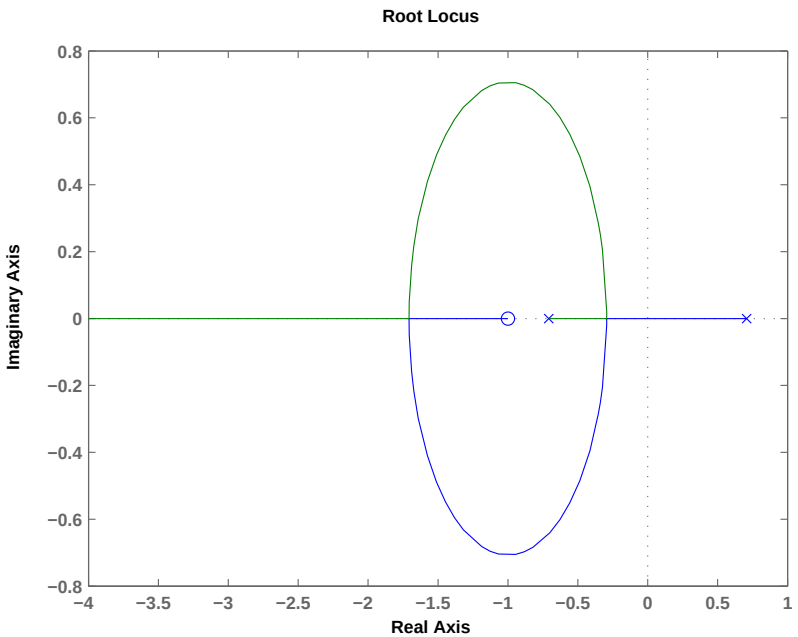


Figure: $K(s) = k(s+1)$

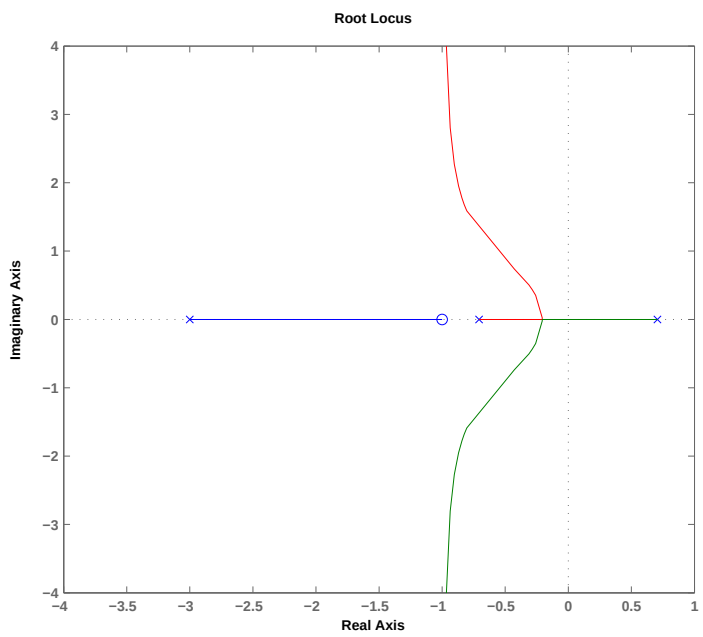


Figure: $K(s) = k\dfrac{s+1}{s+3}$

Lag Compensation

Definition 2.

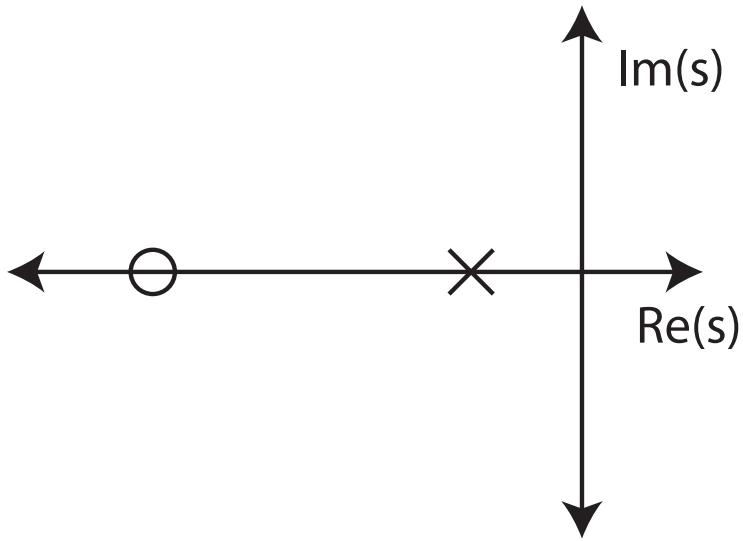
A **Lag Compensator** is a pole-zero compensator

$$K(s) = \frac{s+z}{s+p}$$

where $z < p$.

Used when we really want a pole

- The zero has less effect than the pole.
- Doesn't increase the number of asymptotes.



Lag Compensation

Suspension Problem

$$G(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + s + 1}$$

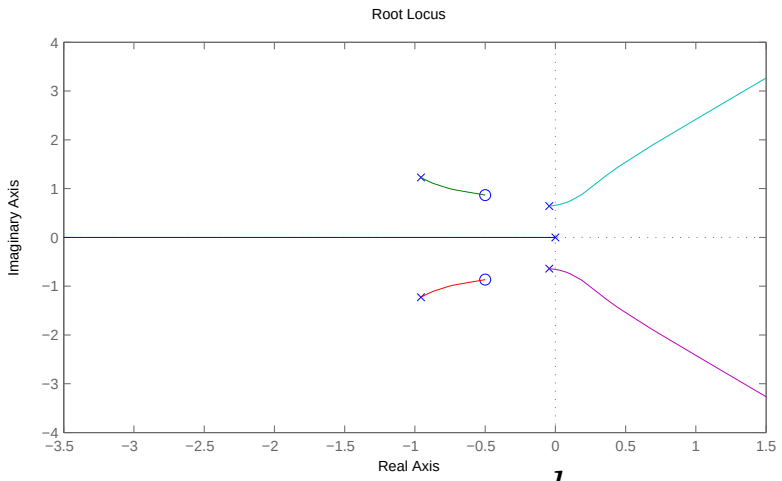


Figure: $K(s) = \dfrac{k}{s}$

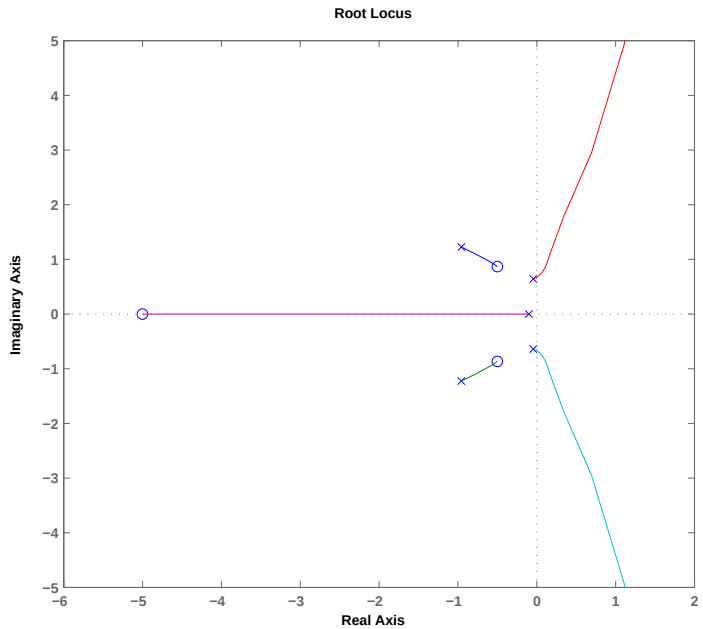


Figure: $K(s) = k\dfrac{s+5}{s+.1}$

Summary

What have we learned today?

Compensation via Root-Locus

- Introduction
- Pole-Zero Compensation
- Lead-Lag

Next Lecture: Pole-Zero Compensation and Notch Filters